

APS110 – Lecture 2

This class is about understanding **SOLID** materials.

Solids can be broken into:

- **Metals**
 - Ex. Al, Fe, Cu...
 - These metals all have properties shared amongst metallic elements
 - Conductive (electrically, thermally)
 - Optically, they tend to be shiny, reflective
 - Mechanically, you can bend them (they can be deformed)
 - Silicon? Is this a metal? It shares some properties with metals, some with ceramics
- **Ceramics**
 - Harder to describe...
 - Chemically, they are often *metal oxides*, ex. Al_2O_3
 - Diamond? (network of carbon atoms in a rigid structure)
 - Properties
 - Hard, brittle
- **Polymers**
 - Ex. “plastics” – polyethylene, polystyrene, Teflon
 - Properties
 - Tend to be non-conductive

Structurally, at the atomic level, ceramics and polymers have drastically different appearances.

Not everything fits neatly into this scheme...

- Semi-conductors
- Graphite – another structure of carbon, with some properties like a ceramic, but with some different properties (structure more like a polymer?)
- Organics – wood is a naturally derived polymer, but behaves like a ceramic in ways (i.e. brittle)
- Wax, butter...soft solids? – malleable, non-conductive...where do they fit?
- Composite materials

We want to start to understand these materials from an engineering perspective.

Mechanical Behaviour

See hand-written notes for drawings.

For most materials, force (F) vs. change in length (Δx) is LINEAR and ELASTIC.

Linear $\rightarrow$ the change in length is proportional to the change in force

- Elastic → when stretched, once the force is released, the material (ex. spring) will return to its original length
- $F = kx$, where **k** is the **spring constant**

The problem with this understanding, is that it *depends on the size of the spring*.

But, when we're comparing different materials, we want something that doesn't depend on the sample size.

→ Need "**Material Property**"

- Material property is *independent* of sample size

We use **STRESS** and **STRAIN**

Hooke's Law

Young's modulus (E) → the constant of proportionality between stress and strain

- Loosely, translates to "stiffness"
- Only describes **elastic** behaviour (does not describe how strong something is)
- Specific to each material
 - Ex. Young's modulus for steel ~ 205GPa